

CYBERSECURITY-LABS

INFORMATION SECURITY

LAB REPORT



Práctica: Clúster HA con Apache

Informe de práctica

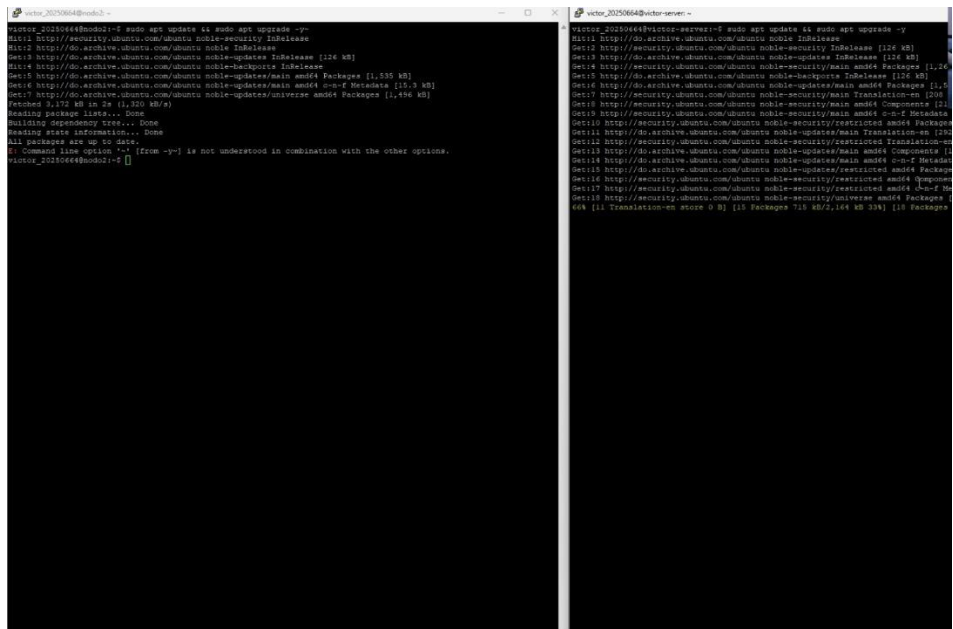
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Information Security

Cybersecurity / Cloud / Networks / GRC

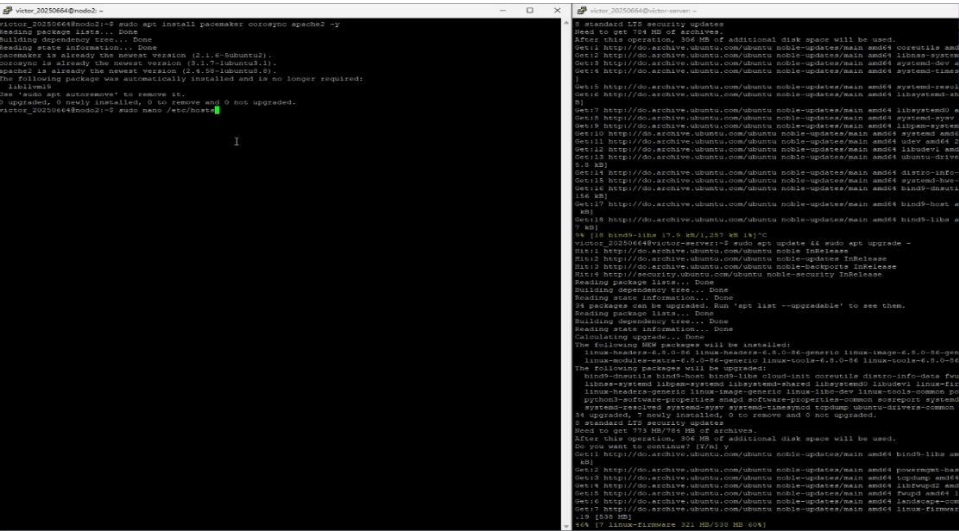


Paso 1: Actualizar los Nodos del Sistema Primero, me aseguro de que mis dos nodos (nodo-1 a la izquierda y nodo-server a la derecha) estén completamente actualizados. Para esto, ejecuto el comando `sudo apt update && sudo apt upgrade -y` en ambos servidores de manera simultánea. De esta forma, confirmo que todos los paquetes de las distribuciones de Ubuntu están en sus últimas versiones estables. Esto es crucial para evitar conflictos con el software del clúster que instalaré más adelante.



The image shows two terminal windows side-by-side. The left window is titled 'victor_2025064@nodo-1' and the right window is titled 'victor_2025064@victor-server'. Both windows display the output of the command `sudo apt update && sudo apt upgrade -y`. The output shows the process of updating package lists and upgrading installed packages. The left window shows updates for packages like 'libnss-systemd', 'libsystemd-shared', 'libsystemd-sd', 'libsystemd-sysv', 'libsystemd-journal', 'libsystemd-udev', 'libsystemd-bash-completion', 'libsystemd-cryptsetup', 'libsystemd-filesystem', 'libsystemd-fsck', 'libsystemd-locale', 'libsystemd-logs', 'libsystemd-networkd', 'libsystemd-notify', 'libsystemd-oomd', 'libsystemd-socket', 'libsystemd-timedate', 'libsystemd-timesyncd', 'libsystemd-userdb', 'libsystemd-xattr', 'libsystemd-zero', 'libsystemd-zerofree', 'libsystemd-zerofree2', 'libsystemd-zerofree3', 'libsystemd-zerofree4', 'libsystemd-zerofree5', 'libsystemd-zerofree6', 'libsystemd-zerofree7', 'libsystemd-zerofree8', 'libsystemd-zerofree9', 'libsystemd-zerofree10', 'libsystemd-zerofree11', 'libsystemd-zerofree12', 'libsystemd-zerofree13', 'libsystemd-zerofree14', 'libsystemd-zerofree15', 'libsystemd-zerofree16', 'libsystemd-zerofree17', 'libsystemd-zerofree18', 'libsystemd-zerofree19', 'libsystemd-zerofree20', 'libsystemd-zerofree21', 'libsystemd-zerofree22', 'libsystemd-zerofree23', 'libsystemd-zerofree24', 'libsystemd-zerofree25', 'libsystemd-zerofree26', 'libsystemd-zerofree27', 'libsystemd-zerofree28', 'libsystemd-zerofree29', 'libsystemd-zerofree30', 'libsystemd-zerofree31', 'libsystemd-zerofree32', 'libsystemd-zerofree33', 'libsystemd-zerofree34', 'libsystemd-zerofree35', 'libsystemd-zerofree36', 'libsystemd-zerofree37', 'libsystemd-zerofree38', 'libsystemd-zerofree39', 'libsystemd-zerofree40', 'libsystemd-zerofree41', 'libsystemd-zerofree42', 'libsystemd-zerofree43', 'libsystemd-zerofree44', 'libsystemd-zerofree45', 'libsystemd-zerofree46', 'libsystemd-zerofree47', 'libsystemd-zerofree48', 'libsystemd-zerofree49', 'libsystemd-zerofree50', 'libsystemd-zerofree51', 'libsystemd-zerofree52', 'libsystemd-zerofree53', 'libsystemd-zerofree54', 'libsystemd-zerofree55', 'libsystemd-zerofree56', 'libsystemd-zerofree57', 'libsystemd-zerofree58', 'libsystemd-zerofree59', 'libsystemd-zerofree60', 'libsystemd-zerofree61', 'libsystemd-zerofree62', 'libsystemd-zerofree63', 'libsystemd-zerofree64', 'libsystemd-zerofree65', 'libsystemd-zerofree66', 'libsystemd-zerofree67', 'libsystemd-zerofree68', 'libsystemd-zerofree69', 'libsystemd-zerofree70', 'libsystemd-zerofree71', 'libsystemd-zerofree72', 'libsystemd-zerofree73', 'libsystemd-zerofree74', 'libsystemd-zerofree75', 'libsystemd-zerofree76', 'libsystemd-zerofree77', 'libsystemd-zerofree78', 'libsystemd-zerofree79', 'libsystemd-zerofree80', 'libsystemd-zerofree81', 'libsystemd-zerofree82', 'libsystemd-zerofree83', 'libsystemd-zerofree84', 'libsystemd-zerofree85', 'libsystemd-zerofree86', 'libsystemd-zerofree87', 'libsystemd-zerofree88', 'libsystemd-zerofree89', 'libsystemd-zerofree90', 'libsystemd-zerofree91', 'libsystemd-zerofree92', 'libsystemd-zerofree93', 'libsystemd-zerofree94', 'libsystemd-zerofree95', 'libsystemd-zerofree96', 'libsystemd-zerofree97', 'libsystemd-zerofree98', 'libsystemd-zerofree99'. The right window shows updates for packages like 'libnss-systemd', 'libsystemd-shared', 'libsystemd-sd', 'libsystemd-sysv', 'libsystemd-journal', 'libsystemd-udev', 'libsystemd-bash-completion', 'libsystemd-cryptsetup', 'libsystemd-filesystem', 'libsystemd-fsck', 'libsystemd-locale', 'libsystemd-logs', 'libsystemd-networkd', 'libsystemd-notify', 'libsystemd-oomd', 'libsystemd-socket', 'libsystemd-timedate', 'libsystemd-timesyncd', 'libsystemd-userdb', 'libsystemd-xattr', 'libsystemd-zero', 'libsystemd-zerofree', 'libsystemd-zerofree2', 'libsystemd-zerofree3', 'libsystemd-zerofree4', 'libsystemd-zerofree5', 'libsystemd-zerofree6', 'libsystemd-zerofree7', 'libsystemd-zerofree8', 'libsystemd-zerofree9', 'libsystemd-zerofree10', 'libsystemd-zerofree11', 'libsystemd-zerofree12', 'libsystemd-zerofree13', 'libsystemd-zerofree14', 'libsystemd-zerofree15', 'libsystemd-zerofree16', 'libsystemd-zerofree17', 'libsystemd-zerofree18', 'libsystemd-zerofree19', 'libsystemd-zerofree20', 'libsystemd-zerofree21', 'libsystemd-zerofree22', 'libsystemd-zerofree23', 'libsystemd-zerofree24', 'libsystemd-zerofree25', 'libsystemd-zerofree26', 'libsystemd-zerofree27', 'libsystemd-zerofree28', 'libsystemd-zerofree29', 'libsystemd-zerofree30', 'libsystemd-zerofree31', 'libsystemd-zerofree32', 'libsystemd-zerofree33', 'libsystemd-zerofree34', 'libsystemd-zerofree35', 'libsystemd-zerofree36', 'libsystemd-zerofree37', 'libsystemd-zerofree38', 'libsystemd-zerofree39', 'libsystemd-zerofree40', 'libsystemd-zerofree41', 'libsystemd-zerofree42', 'libsystemd-zerofree43', 'libsystemd-zerofree44', 'libsystemd-zerofree45', 'libsystemd-zerofree46', 'libsystemd-zerofree47', 'libsystemd-zerofree48', 'libsystemd-zerofree49', 'libsystemd-zerofree50', 'libsystemd-zerofree51', 'libsystemd-zerofree52', 'libsystemd-zerofree53', 'libsystemd-zerofree54', 'libsystemd-zerofree55', 'libsystemd-zerofree56', 'libsystemd-zerofree57', 'libsystemd-zerofree58', 'libsystemd-zerofree59', 'libsystemd-zerofree60', 'libsystemd-zerofree61', 'libsystemd-zerofree62', 'libsystemd-zerofree63', 'libsystemd-zerofree64', 'libsystemd-zerofree65', 'libsystemd-zerofree66', 'libsystemd-zerofree67', 'libsystemd-zerofree68', 'libsystemd-zerofree69', 'libsystemd-zerofree70', 'libsystemd-zerofree71', 'libsystemd-zerofree72', 'libsystemd-zerofree73', 'libsystemd-zerofree74', 'libsystemd-zerofree75', 'libsystemd-zerofree76', 'libsystemd-zerofree77', 'libsystemd-zerofree78', 'libsystemd-zerofree79', 'libsystemd-zerofree80', 'libsystemd-zerofree81', 'libsystemd-zerofree82', 'libsystemd-zerofree83', 'libsystemd-zerofree84', 'libsystemd-zerofree85', 'libsystemd-zerofree86', 'libsystemd-zerofree87', 'libsystemd-zerofree88', 'libsystemd-zerofree89', 'libsystemd-zerofree90', 'libsystemd-zerofree91', 'libsystemd-zerofree92', 'libsystemd-zerofree93', 'libsystemd-zerofree94', 'libsystemd-zerofree95', 'libsystemd-zerofree96', 'libsystemd-zerofree97', 'libsystemd-zerofree98', 'libsystemd-zerofree99'.

Paso 2: Instalación de Paquetes Clave Ahora procedo a instalar los paquetes esenciales de Alta Disponibilidad: `pacemaker`, `corosync` y `apache2` en mis dos nodos. Ejecuto el comando `sudo apt install pacemaker corosync apache2` en ambos terminales para descargar e instalar las últimas versiones. Estos tres componentes son el núcleo de mi clúster, proporcionando el motor de recursos, el mecanismo de comunicación y la aplicación de prueba. ue pondremos en HA.



The image shows two terminal windows side-by-side. The left window is titled 'victor_2025064@nodo-1' and the right window is titled 'victor_2025064@victor-server'. Both windows display the output of the command `sudo apt install pacemaker corosync apache2`. The left window shows the installation of `pacemaker`, `corosync`, and `apache2`. The right window shows the installation of `pacemaker`, `corosync`, and `apache2`. The output shows the process of installing the packages and the resulting system state. The left window shows the installation of `pacemaker`, `corosync`, and `apache2`. The right window shows the installation of `pacemaker`, `corosync`, and `apache2`.

Paso 3: Configurar la Resolución de Nombres (Hosts) Ahora necesito asegurarme de que ambos nodos puedan comunicarse usando nombres en lugar de solo direcciones IP. Para esto, edito el archivo `/etc/hosts` en el nodo izquierdo (nodo-2) usando `sudo nano /etc/hosts`. En ese archivo, agrego las entradas que asocian la IP de cada servidor con su respectivo nombre de host, garantizando que el clúster Corosync pueda identificar correctamente a cada miembro.

```
victor_2025064@node2:~$
GNU nano 7.2 /etc/hosts
127.0.0.1 localhost.localdomain localhost
::1 localhost6.localdomain6 localhost
192.168.68.129 node2
192.168.68.130 node3

# The following lines are desirable for IPv6 capable hosts
::1 localhost ip6-localhost ip6-loopback
fe00::1 ip6-localnet
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts

# standard LFS security updates
Need to get 784 MB of archives.
After this operation, 306 MB of additional disk space will be used.
Get:1 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 coreutils-sys
Get:2 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 libnss-sys
Get:3 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-ev
Get:4 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-tin
1
Get:5 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-co
Get:6 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 libsystemd-
8]
Get:7 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 libsystemd0
Get:8 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-sys
Get:9 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam-sysu
Get:10 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-am
Get:11 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 udev amd64
Get:12 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 libudev1 a
Get:13 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 ubuntu-dri-
6.0 kb]
Get:14 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 disco-inf
Get:15 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-ho
Get:16 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-dnssu
6 kb]
Get:17 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-host
kb]
Get:18 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-liba
7 kb]
# [18 bind9-liba 17.9 kb/1,257 kb 14]C
victor_2025064@victor-server:~$ sudo apt update & sudo apt upgrade -
Hit:1 http://do.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://do.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://do.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
14 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  linux-headers-6.8.0-86 linux-headers-6.8.0-86-generic linux-image-6.8.0-86-g
  linux-modules-extra-6.8.0-86 generic linux-tools-6.8.0-86 linux-tools-6.8.0-8
The following packages will be upgraded:
  bind9-dnssutils bind9-host bind9-lba cloud-init coreutils distro-info-data f
  libnss-systemd libpam-systemd libsystemd-shared libsystemd0 libudev1 linux-f
  linux-headers-generic linux-image-generic linux-libc-dev linux-tool-common
  python3-software-properties smart-software-properties systemd systemd-sys
  systemd-resolved systemd-sysv systemd-timesynod topdown ubuntu-drivers-commo
34 upgraded, 7 newly installed, 0 to remove and 0 not upgraded.
Need to get 773 MB/784 MB of archives.
After this operation, 306 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-liba
kb]
Get:2 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 powermgmt-b
Get:3 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 topdown and
Get:4 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 libfwupd1 a
Get:5 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 fwupd amd64
Get:6 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 landscape-co
Get:7 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-firmw
19 [538 MB]
[1]
```

Paso 4: Asignar el Nombre de Host Correcto Ahora que tengo las entradas en /etc/hosts, debo asegurar que cada máquina reconozca su propio nombre de host. En el nodo izquierdo (mi nodo-2), ejecuto el comando `sudo hostnamectl set-hostname nodo2` para establecer su nombre de manera permanente. Este paso es vital para que Corosync y Pacemaker puedan identificar a cada miembro del clúster por su nombre lógico y único, algo esencial para la comunicación del anillo.

[illegible]

```
victor_2025066@node0-1 ~$ sudo nano /etc/passwd
victor_2025066@node0-1:~$ sudo nano /etc/passwd
victor_2025066@node0-1:~$ sudo hostnamectl set-hostname node0
sudo: Hostnamectl: command not found
victor_2025066@node0-1:~$ sudo systemctl enable pcod apache2
victor_2025066@node0-1:~$ sudo systemctl enable pcod apache2
Synchronizing state of pcod.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable pcod
victor_2025066@node0-1:~$ sudo systemctl start pcod
victor_2025066@node0-1:~$ sudo systemctl status pcod
● pcod.service - PCG GUI and remote configuration interface
   Loaded: loaded (/usr/lib/systemd/system/pcod.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-10-22 23:13:32 UTC; 3h 50min ago
     Docs: manpage(8)
           manpage(8)
Main PID: 7722 (pcsd)
Tasks: 20 (limit: 4544)
Memory: 95.9M [Peak: 112.0M]
CPU: 29.31ms
CGroup: /system.slice/pcod.service
└─7722 /usr/bin/python3 -Es /usr/sbin/pcod
    └─7743 /usr/bin/python3 -Es /usr/sbin/pcod
        └─7753 /usr/bin/python3 -Es /usr/sbin/pcod
            └─7754 /usr/bin/python3 -Es /usr/sbin/pcod
                └─7755 /usr/bin/python3 -Es /usr/sbin/pcod
                    └─7756 /usr/bin/python3 -Es /usr/sbin/pcod
                        └─7758 /usr/bin/python3 -Es /usr/sbin/pcod
                            └─7759 /usr/bin/python3 -Es /usr/sbin/pcod
                                └─7761 /usr/bin/python3 -Es /usr/sbin/pcod
                                    └─7763 /usr/bin/python3 -Es /usr/sbin/pcod
                                        └─7768 /usr/bin/python3 -Es /usr/sbin/pcod
                                            └─7773 /usr/bin/python3 -Es /usr/sbin/pcod

Oct 22 23:39:45 node0 python3[7722]: pam_unix(pcd:auth): authentication failure; logname= uid=0 euid=0 tty=ruser=
Oct 22 23:40:10 node0 python3[7722]: pam_unix(pcd:auth): check pass; user unknown
Oct 22 23:40:20 node0 python3[7722]: pam_unix(pcd:auth): authentication failure; logname= uid=0 euid=0 tty=ruser=
Oct 22 23:50:31 node0 python3[7722]: pam_unix(pcd:auth): authentication failure; logname= uid=0 euid=0 tty=ruser=
Oct 22 23:50:35 node0 python3[7722]: pam_unix(pcd:auth): authentication failure; logname= uid=0 euid=0 tty=ruser=
Oct 22 23:50:39 node0 python3[7722]: pam_unix(pcd:auth): authentication failure; logname= uid=0 euid=0 tty=ruser=
Oct 22 23:50:41 node0 python3[7722]: pam_unix(pcd:auth): authentication failure; logname= uid=0 euid=0 tty=ruser=
Oct 22 23:51:01 node0 python3[7722]: pam_unix(pcd:auth): authentication failure; logname= uid=0 euid=0 tty=ruser=
Oct 22 23:51:17 node0 python3[7722]: pam_unix(pcd:auth): authentication failure; logname= uid=0 euid=0 tty=ruser=
Oct 22 23:51:06 node0 python3[7722]: pam_unix(pcd:auth): authentication failure; logname= uid=0 euid=0 tty=ruser=
lines 1-37/33 (END)
victor_2025066@node0-1:~$ sudo passwd hacluster
New password:
BAD PASSWORD: The password fails the dictionary check - it does not contain enough DIFFERENT characters
retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: No password supplied
retype new password:
No password has been supplied.
passwd: Authentication token manipulation error
passwd: password unchanged
victor_2025066@node0-1:~$ sudo passwd hacluster
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
retype new password:
passwd: password updated successfully
victor_2025066@node0-1:~$ echo "chlo-Servidor Web en HA SO3 Victor Santos. Aqui esta mi prueba! - NODO 2 (129)/hl!" |
sudo tee /var/www/html/index.html
chlo-Servidor Web en HA SO3 Victor Santos. Aqui esta mi prueba! - NODO 2 (129)/hl
victor_2025066@node0-1:~$
```

[illegible]

Paso 7: Configurar la Resolución de Nombres en el Segundo Nodo Ahora, repito el Paso editando el archivo /etc/hosts, pero esta vez en mi nodo de la derecha, el victor-server. Abro el archivo con sudo nano /etc/hosts e ingreso las mismas líneas: la IP y el nombre de host de nodo1 y nodo2. Este paso asegura la simetría en la configuración de la red, permitiendo que ambos nodos puedan resolver los nombres de sus compañeros dentro de la red del clúster.

```
victor_20250664@victor-server: ~  
GNU nano 7.2 /etc/hosts *  
192.168.68.110 nodo1  
192.168.68.129 nodo2  
  
I
```

Paso 8: Habilitar Apache y Configurar hacluster en el Servidor En mi nodo victor-server (Ubuntu Server), primero me aseguro de que el demonio pcsd esté activo y habilitado, lo que me permitirá gestionar el clúster remotamente. Luego, verifico el estado de Apache y, al igual que en el otro nodo, asigno una contraseña segura al usuario hacluster con sudo passwd hacluster. Ambos nodos deben tener la misma contraseña para hacluster para que la autenticación del clúster funcione correctamente.

```
victor_20250664@victor-server: ~  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Pending kernel upgrade!  
Running kernel version:  
6.8.0-85-generic  
Diagnostics:  
The currently running kernel version is not the expected kernel version 6.8.0-  
Restarting the system to load the new kernel will not be handled automatically,  
Restarting services...  
  
Service restarts being deferred:  
/etc/needrestart/nrtest.d/dmns.service  
systemctl restart systemd-logind.service  
systemctl restart unattended-upgrades.service  
  
No containers need to be restarted.  
  
User sessions running outdated binaries:  
victor_20250664 @ session #1: login[643]  
victor_20250664 @ session #5: sshd[2120]  
victor_20250664 @ user manager service: systemd[1160]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
victor_20250664@victor-server:~$ sudo nano /etc/hosts  
victor_20250664@victor-server:~$ sudo systemctl enable pcsd apache2  
Synchronizing state of pcsd.service with SysV service script with /usr/lib/syste  
Executing: /usr/lib/systemd/systemd-sysv-install enable pcsd  
Synchronizing state of apache2.service with SysV service script with /usr/lib/sy  
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2  
victor_20250664@victor-server:~$ sudo systemctl start pcsd  
victor_20250664@victor-server:~$ sudo systemctl status pcsd  
● pcsd.service - PCS GUI and remote configuration interface  
   Loaded: loaded (/usr/lib/systemd/system/pcsd.service; enabled; preset: enab  
   Active: active (running) since Thu 2025-10-23 03:10:38 UTC; 1min 55s ago  
     Docs: man:pcsd(8)  
           man:pcs(8)  
  Main PID: 16640 (pcsd)  
    Tasks: 29 (limit: 4548)  
  Memory: 91.0M (peak: 91.5M)  
     CPU: 1.268s  
   CGroup: /system.slice/pcsd.service  
           └─16640 /usr/bin/python3 -Es /usr/sbin/pcsd  
             └─16647 /usr/bin/python3 -Es /usr/sbin/pcsd  
               └─16657 /usr/bin/python3 -Es /usr/sbin/pcsd  
                 └─16658 /usr/bin/python3 -Es /usr/sbin/pcsd  
                   └─16660 /usr/bin/python3 -Es /usr/sbin/pcsd  
                     └─16664 /usr/bin/python3 -Es /usr/sbin/pcsd  
                       └─16666 /usr/bin/python3 -Es /usr/sbin/pcsd  
                         └─16668 /usr/bin/python3 -Es /usr/sbin/pcsd  
                           └─16675 /usr/bin/python3 -Es /usr/sbin/pcsd  
                             └─16697 /usr/bin/python3 -Es /usr/sbin/pcsd  
                               └─16687 /usr/bin/python3 -Es /usr/sbin/pcsd  
                                 └─16690 /usr/bin/python3 -Es /usr/sbin/pcsd  
  
Oct 23 03:10:37 victor-server systemd[1]: Starting pcsd.service - PCS GUI and re  
Oct 23 03:10:38 victor-server systemd[1]: Started pcsd.service - PCS GUI and re  
victor_20250664@victor-server:~$ sudo passwd hacluster  
New password:  
Retype new password:  
passwd: password updated successfully  
victor_20250664@victor-server:~$
```

Paso 9: Crear la Configuración del Clúster y la Autenticación Primero, modifico la página por defecto de Apache con un mensaje que me permita identificar el nodo. Luego, en el victor-server, utilizo el comando `sudo pcs host auth` para autenticar y configurar mi clúster. Especifico mis nodos (nodo1 y nodo2) y las credenciales del usuario hacluster. Finalmente, utilizo `pcs cluster setup mi_cluster nodo1 nodo2` para crear el clúster con el nombre `mi_cluster` y designar a ambos nodos como miembros.

```
victor_20250664@victor-server: ~
victor_20250664 @ session #1: login[943]
victor_20250664 @ session #5: sshd[2120]
victor_20250664 @ user manager service: systemd[1160]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
victor_20250664@victor-server:~$ sudo nano /etc/hosts
victor_20250664@victor-server:~$ sudo systemctl enable pcsd apache2
Synchronizing state of pcsd.service with SysV service script with /usr/lib/systemd/systemd-sysv-generator
Executing: /usr/lib/systemd/systemd-sysv-generator enable pcsd
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-generator
Executing: /usr/lib/systemd/systemd-sysv-generator enable apache2
victor_20250664@victor-server:~$ sudo systemctl start pcsd
victor_20250664@victor-server:~$ sudo systemctl status pcsd
● pcsd.service - PCS GUI and remote configuration interface
   Loaded: loaded (/usr/lib/systemd/system/pcsd.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-10-23 03:10:38 UTC; 1min 55s ago
     Docs: man:pcsd(8)
           man:pcs(8)
  Main PID: 16640 (pcsd)
    Tasks: 29 (limit: 4548)
  Memory: 91.0M (peak: 91.5M)
    CPU: 1.268s
  CGroup: /system.slice/pcsd.service
          └─16640 /usr/bin/python3 -Es /usr/sbin/pcsd
             16647 /usr/bin/python3 -Es /usr/sbin/pcsd
             16657 /usr/bin/python3 -Es /usr/sbin/pcsd
             16658 /usr/bin/python3 -Es /usr/sbin/pcsd
             16660 /usr/bin/python3 -Es /usr/sbin/pcsd
             16664 /usr/bin/python3 -Es /usr/sbin/pcsd
             16666 /usr/bin/python3 -Es /usr/sbin/pcsd
             16668 /usr/bin/python3 -Es /usr/sbin/pcsd
             16675 /usr/bin/python3 -Es /usr/sbin/pcsd
             16677 /usr/bin/python3 -Es /usr/sbin/pcsd
             16687 /usr/bin/python3 -Es /usr/sbin/pcsd
             16690 /usr/bin/python3 -Es /usr/sbin/pcsd

Oct 23 03:10:37 victor-server systemd[1]: Starting pcsd.service - PCS GUI and remote configuration interface:
Oct 23 03:10:38 victor-server systemd[1]: Started pcsd.service - PCS GUI and remote configuration interface:
victor_20250664@victor-server:~$ sudo passwd hacluster
New password:
Retype new password:
passwd: password updated successfully
victor_20250664@victor-server:~$ echo "<h1>Servidor Web en HA SO3. Victor Santos
ee /var/www/html/index.html
<h1>Servidor Web en HA SO3. Victor Santos Prueba - NODO 1 (132)</h1>
victor_20250664@victor-server:~$ sudo pcs host auth nodo1 -u hacluster -p 0000
nodo1: Authorized
victor_20250664@victor-server:~$ sudo pcs host auth nodo2 -u hacluster -p qwerty
nodo2: Authorized
victor_20250664@victor-server:~$ sudo pcs cluster setup mi_cluster nodo1 nodo2
No addresses specified for host 'nodo1', using 'nodo1'
No addresses specified for host 'nodo2', using 'nodo2'
Error: nodo1: The host seems to be in a cluster already as the following services
, 'pacemaker'. If the host is not part of a cluster, stop the services and retry.
Error: nodo1: The host seems to be in a cluster already as cluster configuration
f the host is not part of a cluster, run 'pcs cluster destroy' on host 'nodo1' to
se --force to override
Error: Some nodes are already in a cluster. Enforcing this will destroy existing
remove the nodes from their clusters instead to keep the clusters working properly
Error: Errors have occurred, therefore pcs is unable to continue
victor_20250664@victor-server:~$ sudo pcs property set stonith-enabled=false
victor_20250664@victor-server:~$ sudo pcs resource create VirtualIP ocf:heartbeat
ask=24 op monitor interval=30s
victor_20250664@victor-server:~$ sudo pcs cluster setup mi_cluster nodo1 nodo2
```


192.168.68.200

No seguro 192.168.68.200

Servidor Web en HA SO3. Victor Santos Prueba - NODO 1 (132)

[illegible]

Paso 13: Confirmación del *Failover* en el Navegador Para la confirmación definitiva, vuelvo a cargar la misma IP virtual (192.168.68.200) en mi navegador web. La página se carga instantáneamente y el mensaje ahora dice: "Servidor Web en HA SO3 Victor Santos. ¡Aquí está mi prueba! - NODO 2". Esto verifica que, después de detener el nodo1, el clúster migró la IP virtual y el servicio web al nodo2 sin interrupción visible para el usuario. La configuración de Alta Disponibilidad ha sido validada con éxito.

